

SUBODH KUMAR

PROFESSIONAL SUMMARY

Applied AI Engineer with 4.5+ years of backend engineering experience and hands-on expertise building intelligent AI-powered applications, Retrieval-Augmented Generation (RAG) systems, conversational AI platforms, and context-aware recommendation engines.

Experienced in designing end-to-end AI workflows including semantic search, vector indexing, embedding pipelines, contextual retrieval systems, voice-enabled AI interactions, and scalable FastAPI backend services. Skilled in bridging backend engineering with Generative AI technologies to develop scalable, production-oriented intelligent systems.

Strong understanding of RAG architecture, FAISS indexing, contextual response generation, model integration, FastAPI APIs, and AI workflow optimization.

CORE EXPERTISE

- Retrieval-Augmented Generation (RAG)
- Conversational AI Systems
- Context-Aware Search & Retrieval
- Semantic Search & Vector Similarity
- FastAPI Backend Development
- AI-Powered Product Engineering
- LLM Integration Workflows
- AI APIs & Model Serving
- Backend Architecture & Scalability
- Predictive Analytics & AI Integration

Applied AI Engineer

RAG Systems | Conversational AI | FastAPI | Vector Search | Python

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PROFESSIONAL EXPERIENCE

Vayam Technologies Ltd.

August 2023 – Present

Product Engineer / Software Engineer

MaddoxPay Fintech Platform

AI Integration & Backend Engineering

- Integrated AI-powered workflows and predictive analytics systems into fintech backend platforms.
- Designed APIs for model serving, intelligent automation, and backend AI integration.
- Worked on deployment-ready AI systems with scalable backend architecture and reliability-focused workflows.
- Built operational analytics systems for anomaly detection, transaction monitoring, and intelligent reporting.

Backend Systems & Product Ownership

- Designed and maintained high-volume payment and recharge systems handling 100K+ monthly transactions.
- Implemented payout systems processing ₹1M+ daily using Axis Bank integrations.
- Developed scalable backend workflows, reconciliation systems, and secure transaction handling.
- Optimized backend performance, database queries, and API response efficiency.

Security & Reliability

- Implemented MFA, geofencing, biometric eKYC integrations, and fraud prevention systems.
- Managed production deployments, uptime monitoring, and infrastructure stability improvements.

Flyde Digital Solutions

July 2022 – August 2023

Fullstack Engineer

- Developed scalable backend systems and integrated multiple third-party APIs.
- Worked on AWS S3-based media handling and modular backend architecture.
- Improved application performance through backend optimization and scalable workflow design.

TECHNICAL SKILLS

Languages & Backend

- Python
- PHP (Laravel)
- SQL
- FastAPI
- REST APIs
- JWT Authentication

Generative AI & NLP

- RAG (Retrieval-Augmented Generation)
- FAISS
- Vector Databases
- Semantic Search
- Context-Aware Retrieval
- Embedding Pipelines
- Conversational AI
- Prompt Engineering
- LLM API Integration
- Voice Assistant APIs

Machine Learning & Analytics

- Scikit-learn
- Pandas
- NumPy
- Feature Engineering
- Predictive Analytics
- Classification & Regression
- EDA
- Model Evaluation

ML Algorithms

- Linear Regression
- Logistic Regression
- Random Forest
- Gradient Boosting
- AdaBoost
- Decision Tree
- XGBoost
- ARIMA/SARIMA

Visualization & Tools

- Streamlit
- Matplotlib
- Seaborn
- Docker
- Git
- Swagger UI

Databases & Infrastructure

- MySQL
- MSSQL
- MariaDB

Saatirmind Software Services

May 2021 – July 2022

Laravel Developer

- Developed backend-focused Laravel applications with scalable APIs and modular architecture.
- Worked on database optimization, backend workflows, and deployment support.

PROJECTS

Smart AI Assistant

Tech Stack: Laravel, Python, FAISS, Vector Database, RAG, NLP, LLM APIs

- Developed an AI-powered intelligent assistant with a custom Retrieval-Augmented Generation (RAG) pipeline for contextual question answering.
- Built complete AI workflows including:
 - Data collection and preprocessing
 - Embedding generation
 - FAISS vector indexing
 - Semantic retrieval pipeline
 - Context-aware response generation
- Implemented vector similarity search using FAISS for efficient semantic retrieval.
- Integrated bilingual conversational support, contextual AI interactions, and intelligent file attachment handling.
- Worked on prompt orchestration, retrieval optimization, response relevance improvement, and scalable AI backend integration.
- Focused on AI system architecture rather than simple API integration.

Career Guidance Engine

Tech Stack: FastAPI, Python, RAG, FAISS, Vector Database, NLP, JWT, Voice APIs

- Built an AI-powered career guidance engine providing personalized recommendations and conversational career assistance.
- Developed a custom RAG pipeline with semantic retrieval and context-aware recommendation workflows.
- Implemented:
 - Context-aware search
 - Vector similarity retrieval
 - Semantic search pipelines
 - Intelligent recommendation systems
- Created vector embeddings and integrated FAISS indexing for efficient contextual information retrieval.
- Integrated voice assistant APIs for voice-enabled AI interactions.
- Designed asynchronous FastAPI backend services with scalable modular architecture.
- Implemented JWT authentication, request validation, and structured AI service workflows.

EDUCATION & CERTIFICATIONS

Executive Post Graduate Certification in Data Science & Artificial Intelligence

Intellipaat | 2025

Microsoft SQL Certification

Intellipaat | 2025

B.Tech – Information Technology

Dr. A.P.J. Abdul Kalam Technical University, Lucknow | 2020

Virtual Lab Development Program (Bootathon)

IIT Bombay | 2018

Industrial Equipment Monitoring & Predictive Maintenance System

Tech Stack: FastAPI, Streamlit, Python, Scikit-learn, Docker

- Engineered an AI-powered predictive maintenance platform for industrial equipment monitoring and fault prediction.
- Built preprocessing and feature engineering pipelines for sensor-based industrial datasets.
- Implemented Random Forest, XGBoost, Logistic Regression, and ensemble learning models for:
 - Fault Classification
 - Predictive Maintenance Forecasting
 - Equipment Failure Detection
- Developed FastAPI backend services and interactive Streamlit dashboards for real-time analytics and explainability.
- Designed scalable Docker-ready architecture for deployment-ready AI systems.

Customer Churn Prediction Platform

Tech Stack: Python, Scikit-learn, Pandas, NumPy, XGBoost

- Developed a machine learning-based customer churn prediction system using multiple classification algorithms.
- Trained and evaluated Logistic Regression, Random Forest, Gradient Boosting, AdaBoost, Decision Tree, and XGBoost models.
- Achieved highest accuracy of 80.3% using Logistic Regression after comparative evaluation.
- Built reusable ML pipelines for preprocessing, evaluation, and prediction workflows.

KEY STRENGTHS

- Generative AI & RAG Systems
- Conversational AI Engineering
- FastAPI & Backend Architecture
- Context-Aware Retrieval Systems
- AI Workflow Engineering
- Semantic Search & Vector Retrieval
- AI Product Development
- Backend + AI Integration
- Scalable Intelligent Systems

HR Analytics & Employee Attrition Prediction

Tech Stack: Python, Pandas, Scikit-learn, Matplotlib, Seaborn

- Developed an HR analytics system to analyze employee attrition patterns and workforce turnover.
- Implemented Logistic Regression and Decision Tree classification models.
- Generated actionable retention insights using EDA, feature engineering, and visualization-driven analysis.

Multimodal Biometric Authentication System

Tech Stack: Python, OpenCV, Computer Vision

- Developed a multimodal biometric authentication system using OpenCV and computer vision techniques.
- Implemented biometric verification workflows with image preprocessing and feature extraction.
- Explored AI-driven authentication and security use cases for real-world applications.